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# A comprehensive system of algorithms for characterization of cardiomyocyte mechanical and electrical function

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ABSTRACT

Understanding the mechanical and electrical function of cardiomyocytes is crucial to unraveling the intricate mechanisms of cardiac physiology and provides insights into the basis of cardiac function and dysfunction. The link between these two aspects provides a holistic perspective and essential knowledge for advancing cardiac research, diagnostics, and therapeutic interventions to combat cardiovascular disease, a leading cause of morbidity and mortality worldwide. A set of Python-based scripts for analyzing mechanical and electrical signals from cardiac cells acquired using Atomic Force Microscopy (AFM) and Multi-Electrode Array (MEA) techniques is presented. The integration of AFM and MEA allows the simultaneous probing of cellular mechanics and electrophysiology. The scripts offer a novel, systematic framework for importing, processing, and interpreting data from complex AFM-MEA experimental setups. Mechanocardiograms obtained from AFM vertical deflection and electrical signals obtained from MEA recordings can be analyzed. The scripts include advanced algorithms for quantifying key parameters, such as beat rate, arrhythmia classification, contraction force, and field potential properties. Additionally, the presented methodology facilitates the correlation between mechanical and electrical behavior. These scripts provide a valuable toolkit for researchers conducting cardiac physiology studies, offering a standardized, efficient approach to unraveling the intricacies of cardiomyocyte behavior *in vitro*.

## 1. Introduction

Cardiac physiology, fundamentally rooted in the mechanics and electrophysiology of cardiac myocytes, has been a central focus in cardiovascular research [1,2]. Understanding the characteristics of these processes is key to deciphering the pathophysiology of cardiac diseases and developing effective therapeutic strategies. Recent advancements in biophysical measurement techniques, particularly those involving Atomic Force Microscopy (AFM) and Micro-Electrode Arrays (MEA), have significantly enhanced our ability to study cardiac cell contraction and electrophysiological properties with unprecedented detail and accuracy [3].

Cardiomyocytes are specialized muscle cells in the myocardium with specific mechanical and electrical functions. Their active subunits are sarcomeres, which contain myosin and actin, the primary proteins involved in contraction, present a unique cell structure [4]. The electrical activity of cardiomyocytes is regulated by ion channels, specifically those for sodium ($Na^+$), potassium ($K^+$), and calcium ($Ca^{2+}$). In muscle cells, including cardiomyocytes, calcium ions are stored and released by the sarcoplasmic reticulum. This plays a crucial role in excitation-contraction coupling [4–6]. Calcium ions bind to troponin, triggering a conformational change that enables myosin heads to bind to actin, leading to filament sliding and muscle contraction [7].

Atomic Force Microscopy (AFM) has emerged as a powerful tool in cardiac research, characterizing properties such as cell growth, phenotype, adhesion, and elasticity [8–10]. AFM can also be used as a sensitive cell-based sensor to measure contractility parameters, such as contraction forces and beat rate, in so-called mechanocardiograms (MCG), making it a powerful tool in heart muscle disease research when combined with patient-derived human pluripotent stem cells differentiated into cardiomyocytes [11]. This technology has been instrumental in various studies, from analyzing the contractile behavior of

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cardiomyocytes to assessing the effects of pharmacological interventions on cardiac cell mechanics. Initially, dissociated human induced pluripotent stem cell-derived cardiomyocytes (iPSC-CMs), harboring a hypertrophic cardiomyopathy mutation, were analyzed. For instance, Liu et al. [12] utilized AFM to evaluate the mechanobiological properties of pluripotent stem cell-derived cardiomyocytes, providing a novel perspective on their contraction force, beat rate, contraction duration, and cellular elasticity. Similarly, Turnbull et al. demonstrated the utility of AFM in characterizing contractility in induced pluripotent stem cell-derived cardiomyocytes (iPSC-CMs), highlighting its potential for cell therapy and drug assessment [13]. AFM can be used to study the effects of various substances on cardiomyocytes [14–17]. AFM is well suited for complex analysis of cell physiology in combination with other methods, for example, using fluorescent dyes and confocal microscopy [3,18,19], Micro-Electrode Array (MEA) [20,21], or even parallel action potential recordings [22].

MEA has been the primary method for understanding the electrophysiological behavior of cardiac cells, enabling screening experiments and drug testing [23,24]. MEAs provide a non-invasive method to record the electrical activity (field potential) of cardiomyocytes that correlates with action potential and its properties [25]. This technology is crucial for evaluating the arrhythmogenic properties of cardiac cells and cardiotoxicity, both of which are vital for healthcare.

The coupling of AFM and MEA technologies presents a groundbreaking approach to cardiac research. By integrating mechanical and electrophysiological analyses, researchers can simultaneously study the contraction forces and electrical activity of cardiac cells. Compared to other patch-clamp approaches, this approach combines well-established AFM and MEA platforms, enabling reproducible, parallel acquisition of mechanical and electrical signals without extensive modifications to existing equipment. [26] Excitation-contraction coupling is a key process in cardiac physiology. Disruption of this fundamental cardiac mechanism underlies many cardiomyopathies. For a more comprehensive overview, readers are directed to the review papers [27,28]. There are other methods that can measure cardiac electrical and mechanical signals individually, but the ability to measure both simultaneously adds significant value to studying these many forms of cardiomyopathies, including catecholaminergic polymorphic ventricular tachycardia, Long QT syndrome 9, Timothy syndrome, and others [3,28]. A notable example of this integrated approach is the work by Saenz Cogollo et al., who combined AFM with a micro-electrode array to study changes in cell topography and the elastic properties of cardiac muscle cells during their contraction-relaxation cycle [21]. This combined technique has proven extremely effective in providing a holistic view of cardiac cell function, enabling a deeper understanding of cardiac physiology and pathology.

Recent advances in synchronous mechano–electrical sensing—particularly the cantilevered micropipette force–electric sensor (CMFES) developed by [29]—demonstrate how a single integrated micropipette can achieve co–localized measurement of nanoscale mechanical deformation and ion–channel currents, including high–resistance sealing and low mechanical/electrical noise, enabling precise quantification of the ~100–130 ms delay between action potential and contraction in cardiomyocytes. While our study uses separate AFM and MEA platforms linked through a unified analytical pipeline rather than an integrated probe, both approaches address the shared challenge of characterizing excitation–contraction coupling. The CMFES represents a powerful single-cell hardware solution, whereas our work provides a scalable, accessible, and algorithm-driven framework suitable for multicellular preparations, high-throughput datasets, and pharmacological screening, thereby complementing these emerging micropipette-based methods.Since the study of the heart using various biosensors is an actively developing area, the question of standardizing data processing methods arises, as well as the development of sufficiently flexible tools that will allow the software to be adapted for various, often related tasks (primary data processing, analysis, and summarization), and also for various objects of study. In this context, we have developed our analysis pipeline in Python, focusing on flexibility, scalability, and reproducibility. The modular architecture of the code enables researchers to easily modify individual components of the workflow, such as filtering, peak detection, or feature extraction, without affecting the overall structure of the algorithm. This design allows seamless adaptation of the same framework to different types of biosignals or measurement techniques, including those beyond cardiac studies. Furthermore, by developing the algorithm in Python, we ensure it remains fully open source and freely available, promoting accessibility and transparency in scientific research. The basic characteristics and logical connections of all the scripts presented here are shown in the diagram in Fig. 1.

The individual scripts focus on describing mechanobiological and electrophysiological data obtained by AFM and MEA. Script's functionality ranges from a simple analysis of beat frequency and signal amplitudes to describing the synchronization of contracting cells, QRST characterization, and a complex description of the relationship between mechanical and electrical signals (mechano-electrical coupling).

## 2. Materials and methods

### 2.1. Data processing overview

In Fig. 2, the overall analytical framework, a comprehensive flowchart summarizing the entire AFM–MEA data processing pipeline. The diagram highlights the two major processing stages used in this work. Stage 1 covers modality-specific operations for both MEA (filtering, peak detection, QRST segmentation, and quality control) and AFM signals (drift correction, contraction cycle detection, and feature extraction). Stage 2 integrates these modalities through precise temporal alignment, beat-to-beat pairing, coupling analysis, and electromechanical decoupling metrics. This overview enables readers to understand how raw synchronized recordings are systematically converted into reproducible quantitative descriptors used for all subsequent analyses.

Unlike existing MEA-only analysis toolkits, the CardioScripts pipeline is modality-agnostic and designed to operate on both MEA field potentials and AFM mechanocardiograms, with a common logic for beat/event detection, synchronized alignment of concurrently recorded streams, and batch-level reproducibility (standardized metadata and exports).

### 2.2. Drug testing

During the experiments, aminophylline (Fagron, Czech Republic), metoprolol (Betaloc, AstraZeneca, UK), isoproterenol (Sigma, Germany), and lidocaine (Sigma, Germany) were used. For aminophylline, a stock solution of 2048 μmol/l was prepared, and then the tested concentrations were obtained by double dilutions. An injection solution containing 1 mg/ml metoprolol tartrate (0.78 mg/ml of metoprolol) was used as a stock solution. The isoproterenol stock solution contained 10 mmol/l of the drug substance. HL-1 (Sigma, Germany) cells were maintained and measured in Claycomb medium as previously described by Klimovic et al. [14]. Embryoid bodies (EBs) were obtained from hESC line CCTL14 (46 XX) derived at Masaryk University, Brno, as previously described [30–32]. All measurements of EBs were done in Tyrode's solution [33]. Stock solutions and intermediate solutions were prepared in the measurement environment.

### 2.3. AFM data acquisition

Embryoid bodies (EBs) obtained from the hESC cell line CCTL14 were cultivated in a sterile Petri dish compatible with the AFM dish heater, which was coated with a gelatin layer to ensure proper adhesion. AFM measurements were carried out on NanoWizard 3 and NanoWizard 4 (Bruker, Germany) using a NitraTall-B cantilever (AppNano, USA).

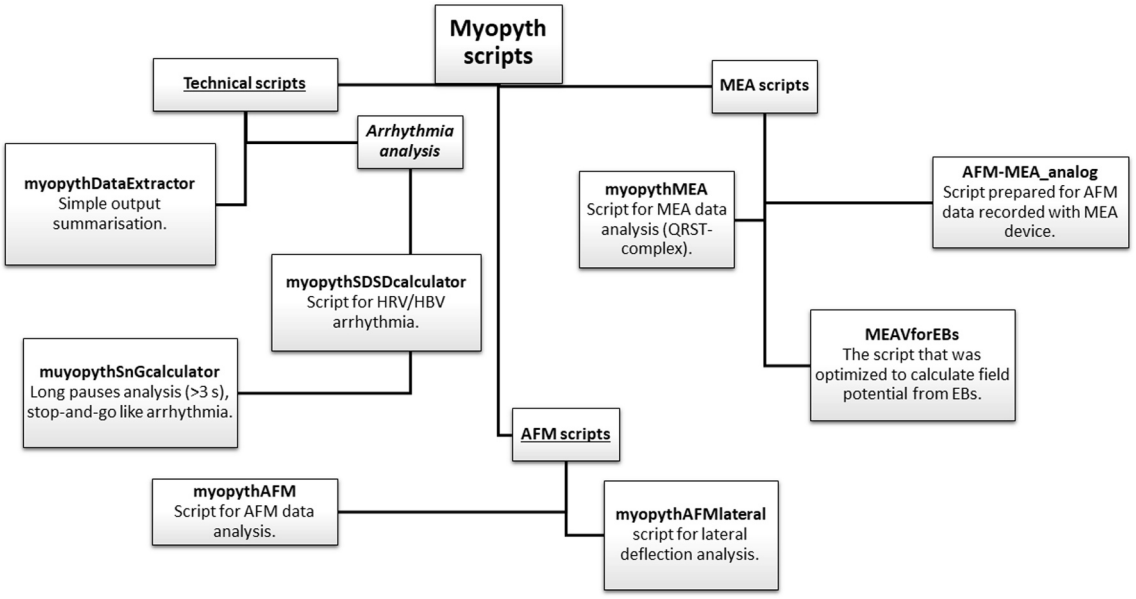



**Fig. 1.** Myopyth family scripts classification. Scripts were prepared to analyze data from cardiomyocytes and embryonic bodies (EBs) obtained using an AFM (mechanocardiogram) and a MEA device (field potential), as well as various technical solutions, including data summation and arrhythmia analysis. Each window displays the script name in bold, followed by a brief description of its function.

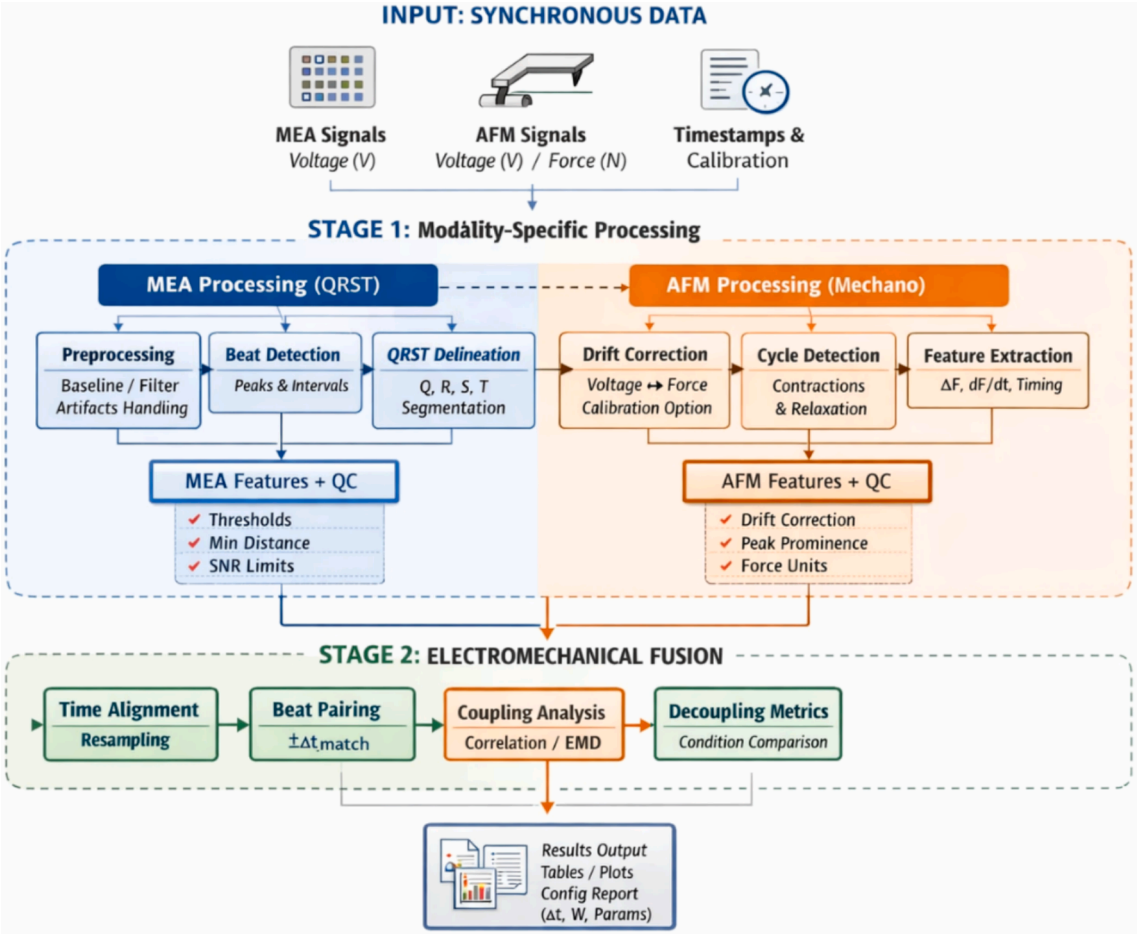



**Fig. 2.** Comprehensive workflow of the AFM–MEA data-processing pipeline. A schematic overview of the complete processing chain used in this study, including (i) modality-specific preprocessing of MEA signals (QRST delineation, beat detection, and field-potential feature extraction) and AFM mechanocardiograms (drift correction, contraction cycle detection, and force-related metrics), followed by (ii) electromechanical fusion through time alignment, beat pairing, coupling analysis, and decoupling metrics. The diagram summarizes how raw synchronous inputs are transformed into the quantitative outputs used throughout the analysis.

Data sets recorded as mechanocardiogram (MCGs), i.e., the force of cluster contractions in real-time [34], were collected at a sampling rate of 100 Hz at 37°C for 10 minutes, with a minimum of 10 minutes after the clusters had stabilized in the presence of treatment. The overall AFM-based measurement setup is shown in Fig. 3, which includes a typical mechanocardiogram (measured force vs. time) curve. The set-point parameter varied between 1 and 2 nN but was maintained throughout the entire measurement. Real-time scan data, including vertical and lateral displacement, were saved and analyzed directly using the myopythAFM script without additional preparation. Measurements were provided accordingly to recommendations given by Pesl et al, describing signal stability [35].

### 2.4. MEA data acquisition

HL-1 cells (Sigma, Germany) were placed on sterile MEA chips with 60 electrodes (Multi Channel Systems, Germany), coated with a mixture of fibronectin and gelatin [14]. Cells were incubated for 3 to 4 days to achieve a high confluence (>90%). For EBs, the incubation time was reduced to 1 day [20]. Measurements were carried out sequentially, starting with a pure medium as a control and then in a medium containing the test substances. Cells were incubated with the treatment for 3 minutes, and then the measurement was performed. Measurements were performed at 10 kHz using a MEA-2100-Mini (Multi Channel Systems, Germany). The data underwent an initial quality check, and data export in the Multichannel Analyzer was performed; no more than half of the channels with a high noise level or without a signal were eliminated (the noise level was determined visually). The whole experimental process, from culturing of cells until the data processing, is schematically shown in Fig. 4.

### 2.5. Simultaneous recording of AFM-MEA data

Data collection to evaluate the effect of electromechanical coupling is performed by combining two previously used experimental approaches: AFM and MEA . EBs are grown on a suitably modified MEA chip surface. This chip was placed in the MEA head stage after achieving perfect cell adhesion, a crucial step for obtaining a sufficiently strong electrophysiological signal from the MEA platform. The MEA chip was then placed under an AFM scanner with an installed cantilever. For this purpose, an inverted optical microscope is successfully used, on whose stage the entire measurement apparatus is mounted. The AFM and MEA data are then collected by a single controller, specifically a multi-electrode array controller. This ensures the signals are timed identically. The signal from the Z-scanner position of the atomic force microscope was transmitted as an analog output by the AFM controller and collected by the MEA controller as an analog input.

To ensure rigorous cross-modal comparability, AFM and MEA streams were preprocessed using modality-appropriate filters and then resampled to a common time base. AFM deflection signals (vertical channel) were low-pass filtered to suppress high-frequency noise and slow drift before peak detection, while MEA field potentials were band-pass filtered with an optional line–notch. Synchrony was enforced by the shared MEA controller clock; for any residual offset, we performed post hoc alignment by maximizing the cross-correlation of beat-locked envelopes within a ±5 ms window. The CardioScripts pipeline was then applied with identical hyper–parameters wherever feasible, and any modality-specific adjustments are reported alongside the corresponding analyses.

### 2.6. MEA data analysis specifications

The script myopythMEA is a comprehensive tool for analyzing MEA-derived field potential data, particularly in cardiac electrophysiology, to identify and visualize peaks associated with the QRST-complex (Fig. 5). It automates filtering, peak detection, and data visualization, enabling efficient, detailed analysis of complex electrophysiological signals. All quality checks were made beforehand using official Multichannel Systems Analyzer software (Multi Channel Systems, Germany). The script was designed to analyze field potential data saved as .CSV files. The code is organized into the following sections:

A. Signal Filtering: The signal filtering stage is designed to suppress high-frequency noise and baseline drift while preserving the morphology of the QRST complex.

The filter_signal function applies to a two-stage zero-phase filtering pipeline:

1. An optional 50 Hz notch filter (IIR notch, Q ≈ 35) removes line interference and other narrowband artifacts.
2. A band-pass Butterworth filter (typically 0.8 – 35 Hz, 6th order) isolates the main physiological frequency band of the cardiac signal.

Both filters are implemented in second-order sections (SOS) form and applied using zero-phase forward-backward filtering (sosfiltfilt) to prevent any phase distortion. This approach offers smoother attenuation than the previous Bessel-based method while maintaining waveform

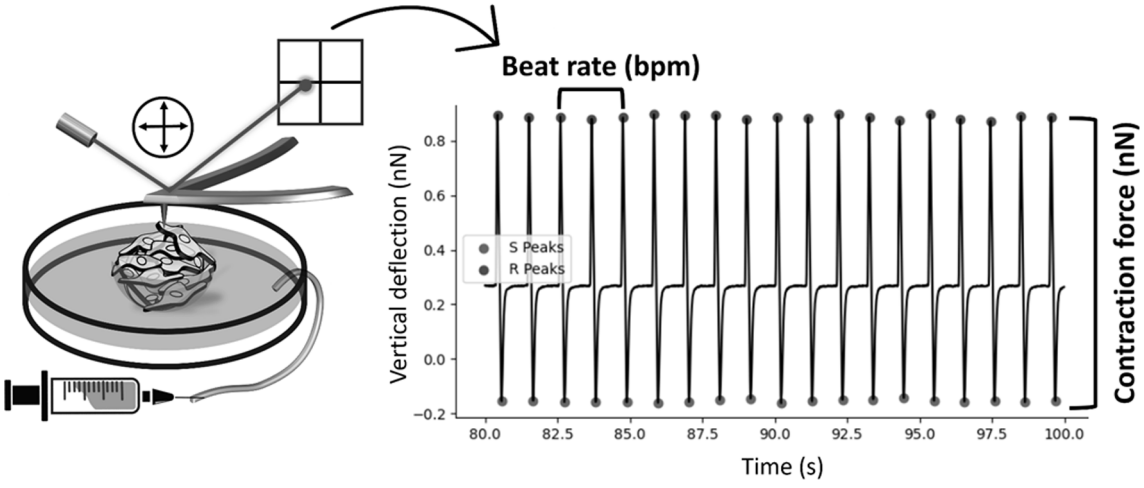



**Fig. 3.** Scheme of AFM experimental setup. *EBs in a petri dish are placed on a heater stage that maintains physiological temperature. The medium is replaced with treatment using an injection syringe. The pre-calibrated cantilever approaches the cells (or a cluster of cells in this case) until it contacts the tip with a specific force, and the detector tracks the cantilever's vertical and lateral displacements. After the experiment, the data will be analyzed using myopyth scripts.*

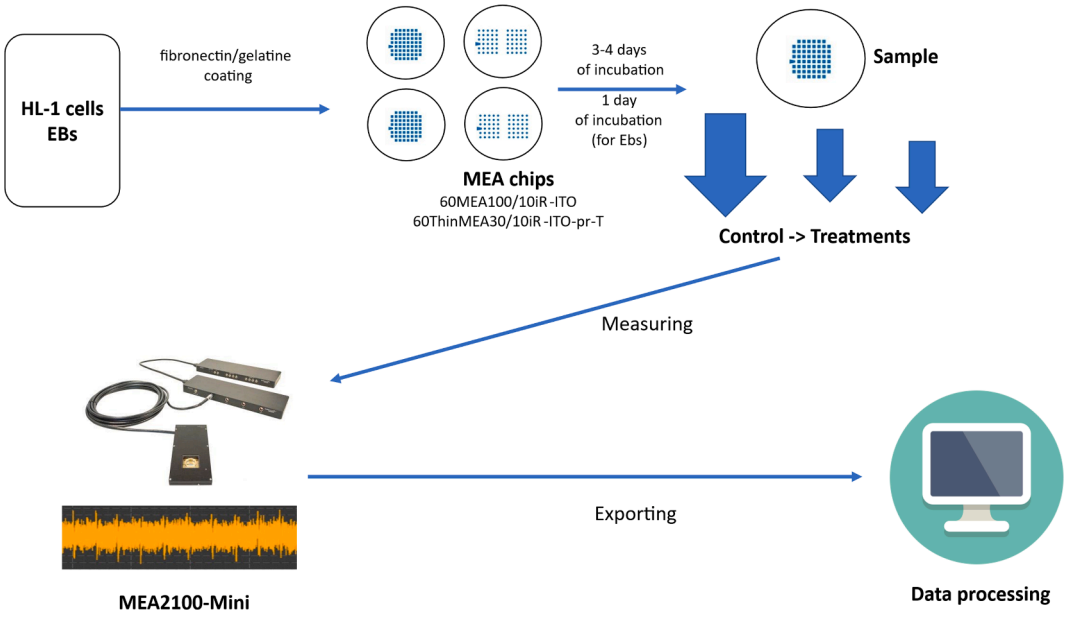



**Fig. 4.** Scheme of MEA experimental setup. The monolayer of HL-1 cells was grown on coated MEA chips. After incubation, measurements were taken for various treatments. After initial processing in the Multichannel analyzer (selection of informative channels), the data was exported and analyzed using myopyth scripts.

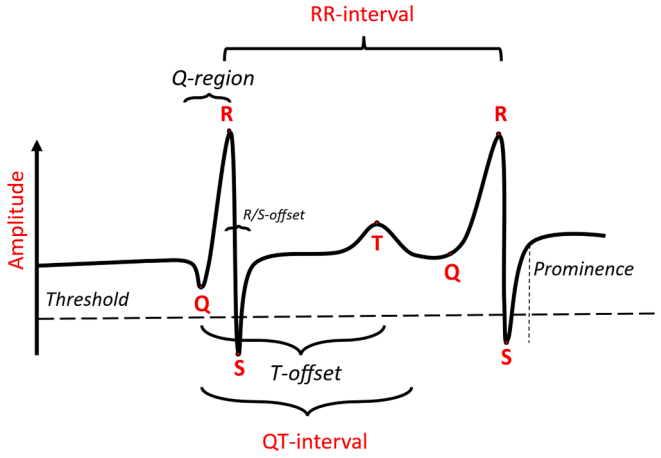



**Fig. 5.** The universal scheme of script settings related to data analysis. The diagram illustrates the functionality of the MyopythMEA script. This tool analyzes field-potential data obtained with an MEA and identifies and visualizes peaks associated with the QRST complex.

fidelity and ensuring accurate detection of QRS and T peaks. Examples of datasets are shown in the Supplementary section.

B. Peak Detection: The find_peaks function is used to identify the negative peaks in the signal, which correspond to the S-peaks in the QRS complex. This is done using the scipy.signal.find_peaks function, with parameters for minimum distance between peaks, height threshold, and prominence, all of which can be adjusted in the settings.

C. QRST Peak Identification: The find_qrst_peaks function detects the characteristic Q, R, and T peaks relative to the previously identified S-peaks. For each S-peak, a local time window is analyzed to locate signal extrema (local minima and maxima). The R-peak is typically the most prominent maximum in the ECG signal and is identified as a local maximum within a short window immediately to the left of the S-peak. The Q-peak is generally the most challenging to detect, but it is usually well-defined and can be found as a local minimum within a small window to the left of the R-peak. The T-peak is detected as a local maximum within the interval between two consecutive S-peaks, corresponding to the repolarization phase of the cardiac cycle. Thus, the function relies on searching for local extrema in adaptive time windows around the S-peaks to robustly identify the Q, R, and T peaks according to their typical physiological locations.

D. Data Plotting: The plot_data function visualizes the entire signal along with the identified Q, R, S, and T peaks. It uses Matplotlib for plotting, displaying both the full signal and segmented views for detailed analysis.

E. Channel Processing: The process_channel function integrates the above steps for each channel in the MEA dataset. It filters the signal, detects S peaks, finds corresponding Q, R, and T peaks, and optionally plots the data.

F. Main Analysis Function: SRfinder orchestrates the analysis across all channels. It scales the time and amplitude data, applies filtering, peak finding, and channel processing, and compiles the results from all channels.

G. Main Execution Logic: Defined in maincode, this section reads the input data file, applies the SRfinder function, and saves the results to an Excel file.

The settings for the analysis, such as header values, scale factors, and peak detection parameters, are adjustable in the settings list. Simplified version for RS peak detection available on GitHub with data extractor (myopathMEAsimple and resultcalc_MEA_old respectively).

### 2.7. Scripts for AFM-based mechanocardiogram data analysis

The Python script myopythAFM is specifically tailored for analyzing cardiac cell contraction data (mechanocardiogram) obtained using Atomic Force Microscopy (AFM). The script performs several critical functions to process and visualize this unique dataset:

Data filtering can be performed and specified (Bessel filter in SciPy library).

To find the S-peak find_peaks function from SciPy library was used. Threshold, prominence, and minimal peak-to-peak distance can be adjusted.

R peaks are identified using a specialized function with the Q-region window parameter (qr).

All datasets will be shown for data quality control. To control the quality of analysis graphs, all the found R and S peaks will be shown.

The main function orchestrates the overall process. It reads the AFM data from a specified file, applies filters for start and end times, previews

the full dataset if enabled, processes the selected channel, and calculates BPM. The function also saves detailed results, including peak (R and S) values and times, BPM per minute, and average BPM, to an Excel file.

The script includes a settings dictionary to configure various parameters such as header row in data, peak detection parameters, and segment length for plotting. The main function is then executed with these settings, along with the specified channel index and time range for analysis.

### 2.8. Technical scripts

A separate version is available for processing AFM lateral channel (myopythAFMlateral) data, featuring the additional function of baseline alignment using linear model fitting from the Scikit-Learn package. Additionally, a version has been prepared for data processing through the initial search for R-peaks (myopythAFM_Rfind).

A separate script (myopythDataExtractor) was prepared for summarizing results from multiple output files generated by the mechanocardiogram analysis script.

myopythSDSDcalculator script is designed to automate the analysis of heart rate/heartbeat variability (HRV/HBV) data with a focus on calculating the Standard Deviation of Successive Differences (SDSD). It operates by first importing HBV data, specifically RR-intervals, from an Excel file. To ensure data reliability, the script checks for any RR-intervals that deviate significantly from the median value, indicating potential outliers (¼ * median < RR-intervals < 4 * median). These outliers are replaced with the median RR-interval to maintain the integrity of the dataset. After ensuring the data quality, the script leverages the pyhrv library to compute the SDSD for each column of RR-interval data. SDSD is a time-domain measure of HBV that reflects the variability in the time intervals between consecutive heartbeats.

myopythSnGcalculator script was designed to process and analyze data from multiple Excel files related to “cut-off” heart rate variability [14]. It works by first creating a support Excel file to store intermediate results. Then, for each input file specified, it loads the data, and for each column, it performs calculations to determine the percentage of values that exceed certain predefined thresholds (hardcoded >3 seconds).

AFM-MEA_analog script has been prepared as a separate technical solution for analyzing AFM data recorded using MEA hardware and software as analog input, retranslated from the AFM controller. MEAVforEBs script allows separate analysis of MEA data obtained from EBs clusters.

### 2.9. Statistical analysis

Experimental data were normalized relative to the control for each sample. The statistical analysis was conducted using GraphPad Prism 8.5 software (GraphPad Software). To identify outliers, the ROUT method was applied with a Q value of 1%. Normality tests were performed on the collected data. For the analysis of hPSC-CM measurements, the Brown–Forsythe ANOVA test, followed by the Games–Howell test for multiple comparisons, was used to assess differences between groups with normally distributed data.

## 3. Results and discussion

In our system, synchronous AFM–MEA validation was performed on EBs, where robust vertical contractions support reliable AFM mechanocardiograms. HL–1 monolayers were used primarily for MEA–based electrophysiology; their motion profile is dominated by lateral displacements with limited vertical deflection, which can reduce AFM mechanical signal reproducibility. The algorithmic pipeline itself is modality–agnostic and operates without modification on either mechanical (AFM) or electrical (MEA) streams, as demonstrated here

### 3.1. Evaluation of myopyth scripts on artificial data

To test the quality of the scripts under controlled conditions, artificial datasets were generated that replicate the main features of data obtained from cardiac cells using MEA (field potential) and AFM (mechanocardiogram). The main signal parameters were known a priori; however, the data analysis was conducted without accounting for these properties to avoid biasing the experiment's outcome. The scripts used to generate the data are available on the project's GitHub (AFM_artificial_data_generator and MEA_artificial_data_generator). Examples of artificial and real datasets are presented in Fig. 6.

The script accurately identified signal peaks and the beat rate, with minimal error, as indicated by the S and R peak accuracy (Table 1).

To assess the script's accuracy in identifying the elements of the QRST complex, we used at least 500 R-R cycles. Specifically, we focused on the mean absolute error (MAE), given in milliseconds, as the most valuable metric with physical meaning. To analyze the script under extreme conditions, we evaluated it on an artificial dataset with additive Gaussian white noise with $\sigma$=5.

Special attention was given to the QT interval, a critical clinical metric [24] correlating with the duration of the action potential. The MAE for the mean QT interval measured by the script was 0.06719 seconds, despite the low accuracy of the exact T-peak indices. It can be described by noise, which causes a slight deformation of the T peak shape.

In data analysis, the choice of tools typically comes down to selecting the optimal solution that balances ease of use, flexibility in settings, and functionality. Myopyth scripts provide tools that enable you to quickly analyze similar data, identify key indicators crucial for cardiological studies, and adapt them to related tasks. The myopyth approach to data, first of all, requires their preliminary unification, since all datasets must first be digitized and saved in text format. Evaluation of a script's performance involves analyzing synthetic data related to cardiac electrophysiology. While the comparison with currently published scientific manuscripts is not specific, the methodology employed and the findings present share similarities with established approaches in the field. The script's accurate peak detection and minimal error in beat rate align with the standards for reliable signal processing algorithms in cardiac research. However, the acknowledged systematic amplitude error, attributed to a low starting sample rate, highlights a common challenge in signal processing: accuracy can be influenced by the quality and resolution of the original data. Use of Q-Q, R-R, S-S, and T-T intervals to assess the accuracy of the script in identifying elements of the QRST complex is the attention given to the QT interval, a critical clinical metric, that demonstrates the script's application in a context relevant to clinical studies [25]. The use of a histogram for graphical representation is a common and effective way to present distribution patterns. The identification of outliers, less than 2% in this case, adds robustness to the evaluation and suggests a high level of consistency in the dataset.

### 3.2. Application of myopyth scripts for data analysis

Studies using AFM as an analytical tool in in vitro cardiology typically employ non-specialized or commonly used data analysis tools. The most commonly used are custom solutions that use MATLAB and related modules [12,34,36]. The same is true for studies that use AFM and MEA together [36,37]. In this case, an option is to search for peaks using two thresholds (for R and S peaks), but this approach has drawbacks for data with insufficiently pronounced peaks or high noise levels. Potentially, multi-operation tasks in which AFM and MEA data could be compared in analysis require additional steps. Also, batch processing becomes significantly more complicated.

Well-known drugs with positive (isoproterenol [38,39]) and with negative (metoprolol [40,41]) chronotropic (beat rate changing) effects were used to evaluate script functionality on real datasets. Using the AFM script for cell clusters mechanocardiogram and the MEA script for

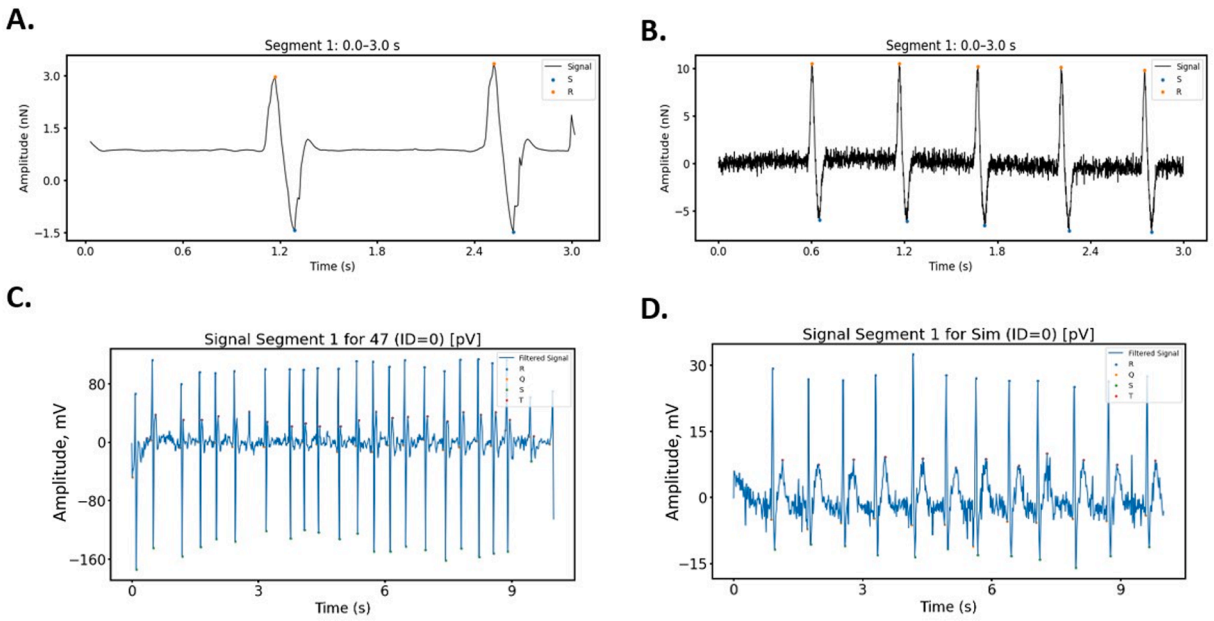



**Fig. 6.** Example of analysis of AFM and MEA datasets using myopyth scripts. Real (A, C) and artificial (B, D) datasets are shown for the AFM and MEA setups, respectively.

**Table 1**
Evaluation of myopyth scripts on artificially generated data.

| Application of myopythAFM script on AFM artificial dataset | | | |
|---|---|---|---|
| Label | MAE | Precision | Recall |
| R | 0.0 | 0.980198 | 1.0 |
| S | 0.0 | 0.980198 | 1.0 |
| **Application of myopythMEA script on MEA artificial dataset** | | | |
| Label | MAE | Precision | Recall |
| Q | 4.955185 | 0.990679 | 0.989362 |
| R | 0.785146 | 1.000000 | 0.998670 |
| S | 2.959228 | 1.000000 | 0.998670 |
| T | 8.598451 | 0.635153 | 0.634371 |
| QT interval | 19.422105 | — | — |

Pre-generated values in an artificial dataset were used for script evaluation. MAE – mean absolute error in Recall and Precision are given as metrics of matching expected and found peaks. Values obtained after script analysis of an artificial dataset are labeled as "received values."

HL-1 cells (immortalized cardiomyocyte cell line), the chronotropic effect was proved (Fig. 7). The inotropic effect, i.e., an increase in contraction amplitude, was verified using aminophylline [42–44]. A significant ($p < 0.01$) positive inotropic effect was shown.

In the experiment shown in Fig. 3, the functionality of computational scripts designed to quantify cardiac arrhythmia in an *in vitro* setting was investigated. The experiment's methodology involved treating EBs with aminophylline to induce arrhythmic patterns, which could be measured using atomic force microscopy (AFM). An exact methodology and a physiological explanation were described in depth in [14]. Processed data examples can be found in the Supplementary section. Raw datasets are available at the full data repository

Determining arrhythmia based on the distances between R-R intervals is a useful parameter in cardiologic research [45,46]. Panel A (Fig. 8A) highlights the incidence of arrhythmias, specifically those with beat-to-beat intervals greater than 3 seconds, which are considered indicative of fibrillation-like arrhythmia. This classification is supported by the findings of Klimovic et al. [14], who identified similar patterns as characteristic of fibrillation. To ensure the robustness of our findings and to demonstrate the consistency of different methodological approaches, we further analyzed the data using the Standard Deviation of Successive Differences (SDSD) method [47,48], as shown in Panel B (Fig. 8B). The SDSD is a measure of the variability in the time intervals between consecutive heartbeats and is a commonly used metric in the study of heart rate variability and arrhythmia. The measurements revealed increased variability in heartbeat intervals in the aminophylline-treated EBs.

One of the key advantages of myopyth scripts for analyzing AFM data is the ability to use them for various channels, including lateral deflection and height, as well as the analysis of smaller peaks, for example, when searching for additional events originating from smaller cell clusters that are in contact with the cantilever. One disadvantage is the need to fine-tune the script before use, the difficulty of identifying the necessary fragments in the data, and the lack of a system that allows switching between solutions for different types of data.

Aminophylline tends to prolong QT intervals with increasing beat rate [42,49]. The ability of aminophylline at a concentration of 128 μM to alter the duration of QT intervals in HL-1 cardiomyocytes was experimentally studied (represented in Fig. 9) using myopyth scripts. A significant ($p < 0.05$) increase in QT-interval duration was observed, accompanied by a fairly large dispersion of the data. Therefore, it is recommended to evaluate the corresponding effects using larger data samples.

To test the MEA script and validate the AFM-MEA contraction coupling system, an experiment was conducted on EBs using lidocaine, which is known to cause decoupling in the electromechanical signal [50]. It was shown (Fig. 10) that lidocaine at concentrations of 25 and 30 μM significantly ($p < 0.001$) altered the distance between the electrical signal and the mechanical contraction of the cells compared to the control in a dose-dependent manner.

Signal processing solutions for MEA can vary significantly from one another. As with AFM, conventional data processing tools such as Origin software can be used [47]. This can be inconvenient when analyzing a large dataset or scaling up studies.

Cardio2D software [51], developed by Multi Channel Systems, is a specialized tool primarily designed for cardiac research and studies on cardiotoxicity. However, this solution has low flexibility and can only analyze data recorded using the accompanying data acquisition tool. In addition, like some other tools [52,53], Cardio 2D focuses on local activation times and excitation mapping, rather than the arrhythmic

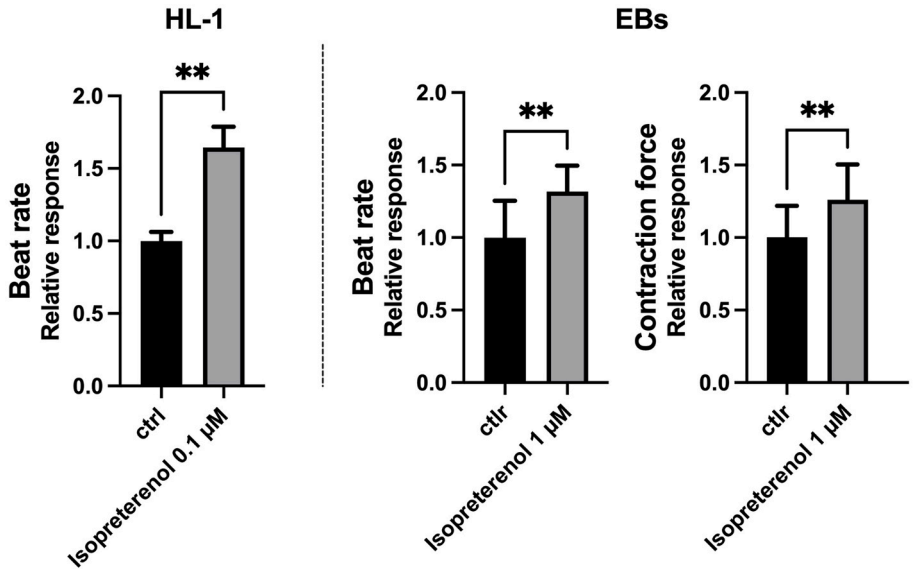



**Fig. 7.** Effect of isoprotenerol on beating rate and contraction force on HL-1 cells and EBs. (Beat rate values were obtained with MEA and AFM measurements on HL-1 (N = 5 for control and treatment) cells and EBs (N = 10 for control and treatment), respectively. Contraction force was obtained with AFM on EBs (N = 10). In all measurements, significant effects ($p < 0.05$ marked as one asterisk, $p < 0.01$ marked as two asterisks) were found. All values were normalized to the control (taken as 1.0).

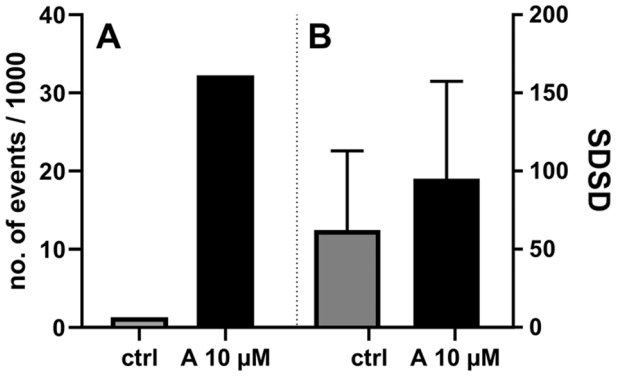



**Fig. 8.** Arrhythmia analysis of EBs (N = 10) measured via AFM. Panel A illustrates the frequency of arrhythmic events characterized by beat-to-beat intervals exceeding 3 s per 1000 events. The control group (ctrl) demonstrates a baseline frequency of arrhythmic events, whereas the group treated with aminophylline at 10 μM (A 10 μM) shows an increased frequency of such events. An exact methodology and a physiological explanation were described in depth in [14]. Panel B represents the standard deviation of successive differences (SDSD) in heartbeat intervals as a measure of beat rate variability arrhythmia. Here, the control group (ctrl) establishes the baseline beat rate variability, with the aminophylline-treated group (A 10 μM) exhibiting a slight increase in beat rate variability. Error bars indicate standard deviation, highlighting the variability within each group. Examples of processed data can be found in the Supplementary. Raw datasets are available at the full data repository.

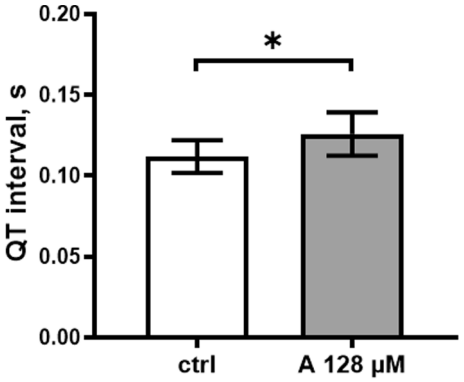



**Fig. 9.** Effect of aminophylline with concentration 128 μM (A 128 μM; N = 891) compared with control (ctrl; N = 782) on QT-interval duration in HL-1 cells. Significance ($p < 0.05$) is marked with one asterisk. Values were subtracted by 0.2 as the script MAE and normalized to R-R according to the interval, exact methodology, and physiological explanation, as described in depth in [14]. Examples of processed data can be found in the Supplementary. Raw datasets are available at the full data repository.

properties of cells; however, this does not exclude the joint use of this program and myopyth scripts to carry out a more complex analysis.

There are solutions [54,55] designed for high-throughput analysis of data obtained using multiwell MEA. Despite the ability to scale and adapt myopyth scripts, they lack an automatic system for finding optimal parameters, require partial control of settings, and are also not optimized for large volumes of data. Currently, this is an open task to enhance the product, as well as to introduce other algorithms for arrhythmia analysis and increase the information content of the analysis in cases where MEA is correlated with other methods.

We declare that there are currently no alternative complete systems for the comprehensive assessment of cardiomyocyte physiological parameters using the AFM-MEA setup as an experimental system. In addition, a system is proposed for analyzing several key parameters in these signals, primarily related to arrhythmia and expressiveness of contractions.

### 3.3. Cross–modal validation on synchronous AFM–MEA recordings (EBs)

We evaluated the modality-agnostic behavior of CardioScripts on EB datasets acquired synchronously with AFM and MEA. As shown in Fig. 10, the pipeline detects events in both modalities and quantifies the electromechanical timing relationship. Pharmacological lidocaine challenge increased the delay between the electrical field potential and mechanical contraction in a dose-dependent manner (Fig. 10C), consistent with electromechanical decoupling. This demonstrates that the same analysis strategy can be applied to concurrent mechanical and electrical signals and is sensitive to expected physiological

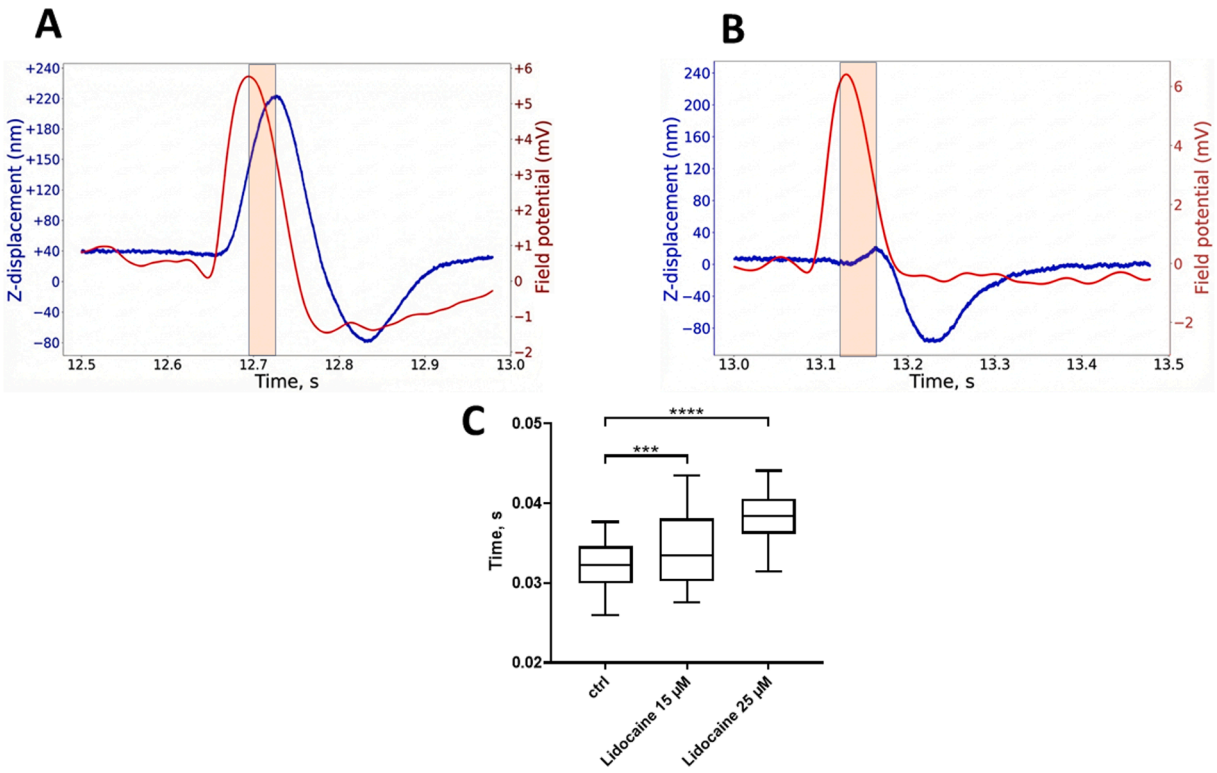



**Fig. 10.** Simultaneous measurements Representative peaks of mechanical (mechanocardiogram, blue line) and electrical (field potential, red line) contractions of EBs in control (A) and with 25 μM of lidocaine (B). The baseline was subtracted from zero. Measured simultaneously. The y-axis was normalized on the R-peak, the field potential was increased 100 times, and the AFM signal was decreased 5 times. The orange rectangle marks the displacement between signals. (C) Distances (decoupling) between electrical signal events on MEA and mechanical signal events from AFM in EBs in control measurement (ctrl; N = 423) and with lidocaine in concentrations 15 μM and 25 μM (N = 63 and N = 192, respectively). Significance $p < 0.001$ marked as three asterisks, $p < 0.0001$ marked as four asterisks.

perturbations.

Comparison to micropipette-based synchronous systems. In contrast to cantilevered micropipette force–electric sensors (CMFES), which combine a flexible glass micropipette for force readout with a patch–-style electrical interface at the single–cell level, our synchronous platform couples AFM vertical–deflection mechanocardiograms with MEA field potentials on embryoid body (EB) preparations, with simultaneity enforced by a shared MEA controller clock and optional post–hoc cross–correlation alignment within ±5 ms (Methods Sec. 2.4). CMFES intrinsically co–localizes mechanical and electrical signals in one probe and has been used to resolve the electromechanical delay on the order of ~100–130 ms in single cardiomyocytes; our EB datasets likewise capture the AFM–MEA timing relationship and reveal dose–dependent increases in delay under lidocaine, demonstrating sensitivity to electromechanical decoupling at the tissue–scale (Fig. 10). While micropipette force sensors can attain piconewton–level force sensitivity and millisecond temporal resolution, they typically operate at low throughput and require skilled manipulation; by contrast, our AFM–MEA scheme (AFM sampled at 100 Hz; MEA at 10 kHz) is compatible with multicellular preparations and batch pharmacology (e.g., isoproterenol, metoprolol, aminophylline, lidocaine), enabling population–level metrics with a single analysis pipeline. Taken together, CMFES provides an intracellular, high–precision single–cell benchmark for excitation–contraction delay, whereas our approach emphasizes a scalable, modality–agnostic workflow applicable to multicellular contexts and drug–response profiling; the two are technically complementary and address different experimental scales and use–cases.

Positioning relative to integrated single–cell probes. Micropipette–based "cantilevered" sensors integrate force and electrical measurements in one probe and have been used to quantify cardiomyocyte electromechanical timing at the single–cell level. In contrast, CardioScripts emphasizes a software-driven, platform-agnostic workflow for AFM–MEA datasets on multicellular preparations, enabling batch pharmacology and modality-consistent analytics; these approaches are technically complementary. [29].

### 3.4. Limitations

Our synchronous validation was conducted in EB preparations rather than in HL-1 monolayers, given the latter's predominantly lateral motion and less reproducible vertical AFM signal. Future work will extend synchronous measurements to additional preparations (including HL–1 under optimized conditions) and expand quantitative analyses of the electromechanical timing relationship.

## Authors contribution

DK prepared the scripts, processed the data and wrote the main text of the article. DK and SK carried out the main measurements and obtained the initial data. DB prepared the cells and EBs. VR and MP participated in the discussion of the results and supervised the experiments. JP created the article concept and experimental design, provided guidance in the preparation of scripts, article, and analysis of results. All authors read the text of the article and took part in the discussion and corrections.

## CRediT authorship contribution statement

**Daniil A. Kabanov:** Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Simon Vrana:** Cell measurement, Data Curation, Statistical evaluation. **Deborah Beckerová:** Cell cultivation, CMs differentiation, Methodology. **Vladimir Rotrekl:** Resources. **Martin Pesl:** Resources, Project administration, Data curation, Conceptualization. **Jan Přibyl:** Writing – original draft, Resources,

Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bspc.2026.110125.

## Data availability

All scripts are available on GitHub (https://github.com/DaniilKabanov/CardioScripts). Experimental data are available in the Zenodo repository (https://zenodo.org/records/17523923).

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